Uses

1. Used in electronic laboratories.
2. Used to test and align electronics equipment.
3. Used to rectify faults in Radio and television receivers.
4. It can produce signals having frequencies from 0.01 Hz to 100 MHz.

10.6. SPECTRUM ANALYSER

An instrument, which can analyse the parameters (frequency, phase etc.,) of a signal, is known as spectrum analyser.

Signals can be seen in the display of the spectrum analyser that can be analysed by the user. It works on the basis of super heterodyning principle. Figure 10.26 shows the front panel of a spectrum analyser.

10.6.1. Functions

A Spectrum Analyser can analyse

- Time domain signals
- Frequency domain signals and
- Modulation of signals

Time Domain Analysis

- Changes in time of a signal can be seen

Frequency Domain Analysis

- Each frequency component of a complex signal can be separated easily
- Low level distortion signals can be detected
- Spurious signal can be found out

Modulation of signals - Analysis

- Changes in frequency can be seen
- The modulation accuracy can be analysed but changes in amplitude cannot be seen

10.6.2 Applications

It is used for signal analysis in:

1. Mobile phone transmitters
2. Satellite communication
3. T.V transmitters and receivers
4. Cable TV broadcasting
5. Internet broadband transmission and reception

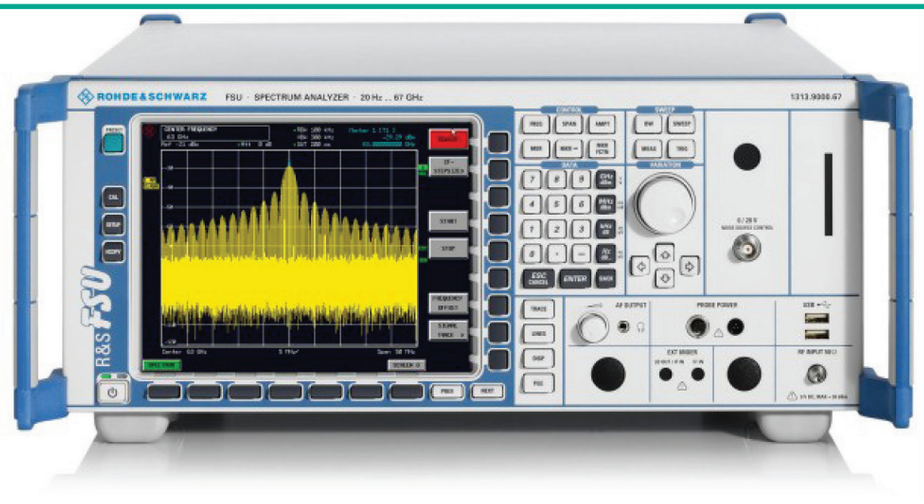


Figure 10.26 Front panel of a Spectrum Analyser

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